Two-Stage Teaching Assignment Model

Linear optimization structure with integer teaching-group decisions and scenario-based recourse

1. Model purpose

This model provides a simple and operational two-stage stochastic teaching assignment framework for course scheduling under uncertain demand. In the first stage, the school prepares a baseline teaching plan before the actual number of course groups is known. In the second stage, once actual demand is realized, additional adjunct capacity is used to cover any unmet teaching demand.

The formulation is intentionally compact so that it remains easy to implement, explain, and align with the PuLP code. Compared with a highly detailed timetable model, this version is more suitable for coursework and presentation.

2. Main assumptions

- Each course has a known minimum and maximum number of groups.
- The first-stage plan covers the minimum required groups for each course.
- All regular faculty members must be used in the first stage through teaching-load bounds.
- Adjunct instructors are already allowed in the first stage because regular faculty alone may not cover all baseline demand.
- In the second stage, only extra adjunct teaching is used as the recourse action.
- Faculty preference is treated as a soft constraint through a penalty in the objective function.
- When no detailed probability distribution is available, actual demand is generated by Monte Carlo simulation using a discrete uniform distribution.
- All teaching-group allocation variables are integers because fractional groups are not meaningful in practice.

3. Sets and indices

Symbol	Meaning
C	Set of courses
F	Set of regular faculty members
S	Set of demand scenarios
$c \in C$	Course index
$f \in F$	Faculty index
$s \in S$	Scenario index

4. Parameters

Parameter	Definition
cred_c	Teaching credits per group of course c

g_c^min	Minimum number of groups required for course c
g_c^max	Maximum possible number of groups for course c
L_f	Minimum teaching load of faculty member f
U_f	Maximum teaching load of faculty member f
p_cf	Preference score of faculty member f for course c, typically from 0 to 5
α	Unit cost of adjunct teaching in the first stage
β	Unit cost of additional adjunct teaching in the second stage
λ	Weight of the faculty preference penalty
g_cs	Realized number of groups for course c under scenario s

Demand values are generated from the course-level interval:

$$g_{cs} \in \{g_c^{\min}, g_c^{\min} + 1, \ \dots, g_c^{\max}\}$$

5. Decision variables

First-stage variables

- y_cf: number of groups of course c assigned to faculty member f in the first stage.
- a_c: number of groups of course c assigned to adjunct instructors in the first stage.

Second-stage variables

- u_cs: additional adjunct groups assigned to course c under scenario s.

All decision variables are nonnegative integers:

$$y_{cf} \in \mathbb{Z}_+, \quad a_c \in \mathbb{Z}_+, \quad u_{cs} \in \mathbb{Z}_+$$

6. Objective function

The total objective combines three parts: first-stage adjunct cost, faculty preference penalty, and expected second-stage recourse cost.

$$\min \alpha \sum_{c \in C} a_c + \lambda \sum_{c \in C} \sum_{f \in F} (5 - p_{cf}) y_{cf} + \frac{1}{|S|} \sum_{s \in S} \beta \sum_{c \in C} u_{cs}$$

Interpretation

- The first term reduces baseline reliance on adjunct instructors.
- The second term penalizes assigning courses to faculty who have lower preference for them.
- The third term minimizes the expected cost of emergency teaching adjustment in the second stage.

7. Constraints

7.1 First-stage baseline demand coverage

For each course, the first-stage assignment must cover the minimum required number of groups.

$$\sum_{f \in F} y_{cf} + a_c = g_c^{\min}, \quad \forall c \in C$$

This makes the baseline plan a conservative schedule built around guaranteed demand.

7.2 Faculty teaching-load constraints

Each regular faculty member must be assigned a workload between the minimum and maximum allowable teaching loads.

$$L_f \leq \sum_{c \in C} \text{cred}_c \, y_{cf} \leq U_f, \quad \forall f \in F$$

These bounds ensure that every faculty member is used while no one is overloaded.

7.3 Second-stage recourse constraints

Once scenario s is realized, the baseline plan plus the extra adjunct assignment must be sufficient to cover actual demand.

$$\sum_{f \in F} y_{cf} + a_c + u_{cs} \geq g_{cs}, \quad \forall c \in C, \ \forall s \in S$$

If realized demand exceeds the baseline plan, additional adjunct groups are added as recourse.

7.4 Integrality and nonnegativity

$$y_{cf}, \ a_c, \ u_{cs} \in \mathbb{Z}_+$$

These conditions are necessary because course groups are discrete units.

8. Scenario generation by Monte Carlo simulation

To represent uncertain future teaching demand, the model uses Monte Carlo simulation. For each scenario s and each course c, the realized number of groups is drawn from all integer values between the minimum and maximum group numbers.

$$g_{cs} \sim \text{Discrete Uniform}(g_c^{\min}, g_c^{\max})$$

- Typical settings are 50 or 100 scenarios.
- A fixed random seed, such as 42, is recommended for reproducibility.
- This approach is reasonable when only lower and upper bounds are available and no historical probability distribution is given.

9. Full Model

Putting everything together, the full model is:

$$\min \ \alpha \sum_{c \in C} a_c + \lambda \sum_{c \in C} \sum_{f \in F} (5 - p_{cf}) y_{cf} + \frac{1}{|S|} \sum_{s \in S} \beta \sum_{c \in C} u_{cs}$$

subject to

$$\sum_{f \in F} y_{cf} + a_c = g_c^{\min}, \quad \forall c \in C$$

$$L_f \leq \sum_{c \in C} \text{cred}_c \, y_{cf} \leq U_f, \quad \forall f \in F$$

$$\sum_{f \in F} y_{cf} + a_c + u_{cs} \geq g_{cs}, \quad \forall c \in C, \ \forall s \in S$$

$$y_{cf}, a_c, u_{cs} \in \mathbb{Z}_+, \quad \forall c, f, s$$